

5.5.2 Substitution w/ Definite Integrals Worksheet

Evaluate each of the following integrals.

1. $\int_0^1 \sqrt[3]{1+7x} dx$

2. $\int_0^{\pi} \sec^2\left(\frac{t}{4}\right) dt$

3. $\int_0^{13} \frac{dx}{\sqrt[3]{(1+2x)^2}}$

$$4. \int_1^2 x\sqrt{x-1}dx$$

$$5. \int_{1/2}^1 \frac{\cos(x^{-2})}{x^3} dx$$

$$6. \int_0^{\sqrt{\pi}} x \cos(x^2) dx$$

$$7. \int_{1/6}^{1/2} \csc(\pi t) \cot(\pi t) dt$$

$$8. \int_0^{\pi/2} \cos(x) \sin(\sin(x)) dx$$

$$9. \int_0^4 \frac{x}{\sqrt{1+2x}} dx$$

$$10. \int_e^{e^4} \frac{dx}{x\sqrt{\ln(x)}}$$

$$11. \int_0^1 \frac{e^z + 1}{e^z + z} dz$$

$$12. \int_0^1 xe^{-x^2} dx$$

13. $\int_0^{1/2} \frac{\sin^{-1}(x)}{\sqrt{1-x^2}} dx$

14. $\int_0^1 \frac{dx}{(1+\sqrt{x})^4}$. *Hint: Try to rewrite your du to be fully in terms of u instead of x.*

#15 – 18 Use properties of even and odd functions to help answer the following questions.

15. Evaluate $\int_{-\pi/3}^{\pi/3} x^4 \sin(x) dx$.

16. Evaluate $\int_{-1}^1 \left(x^5 - 6x^9 + \frac{\sin(x)}{(1+x^4)^2} \right) dx.$

17. Evaluate $\int_{-1}^1 \frac{\tan(x)}{1+x^2} dx.$

18. Evaluate $\int_{-\pi/4}^{\pi/4} \frac{t^4 \sin(t)}{2 + \cos(t)} dt.$

19. Suppose f is an odd function and g is an even function, each integrable on $[-2, 2]$. Give that

$$\int_0^2 f(x) dx = 3, \quad \int_2^6 f(x) dx = 13 \text{ and } \int_0^2 g(x) dx = 6, \text{ evaluate the following.}$$

a) $\int_{-2}^2 [4f(x) - 5g(x)] dx$

b) $\int_{-2}^2 [3g(x) + x^3 - 7] dx$

c) $\int_{-2}^2 [f(x)g(x)] dx$

d) $\int_{-2}^0 f(x) dx$

e) $\int_{-2}^2 \left[(f(x))^3 + 4 \right] dx$

f) $\int_{-6}^2 \left[f(x) + 2 \right] dx$

20. If $f'(x)$ is continuous on $[a, b]$, show that $2 \int_a^b f(x) f'(x) dx = [f(b)]^2 - [f(a)]^2$.

21. If $f(x)$ is continuous on $[0, 1]$, prove that $\int_0^1 f(x) dx = \int_0^1 f(1-x) dx$.

22. If $f(x)$ is continuous and $\int_0^4 f(x) dx = 10$, find $\int_0^2 f(2x) dx$.

23. If $f(x)$ is continuous and $\int_0^9 f(x) dx = 4$, find $\int_0^3 xf(x^2) dx$.